Preview: set-01/by_level/Grade_03-04_Ecolier/y2023_q22 (v1)

Turning machine and printing machine: the corner mark counts all the turns, the picture only the turns after it was printed

How to think about it: a turning machine and a printing machine

Machine R turns the sheet a quarter turn clockwise. Machine S prints an upright club. Two things on the sheet tell you what happened:

- **The corner mark** is there from the start. It turns with **every** R. So it tells you how many times R was used in total.
- **The club** is printed upright. It only turns with the R machines **after** S.

The contest question

The mark went from bottom left to top right: 2 quarter turns, so R is used twice. The club lies on its side (stem to the left): 1 quarter turn, so one R comes after S. Order: R, S, R.

Work backwards: the club shows the turns after printing, the mark shows all the turns.

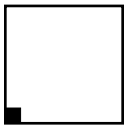
Watch out

- Four quarter turns bring the sheet back to where it started.
- If you only look at the club, you miss the turns before S. If you only look at the mark, you do not know where S goes.

Try it: draw a little square in a corner of a sticky note and turn it in your hand.

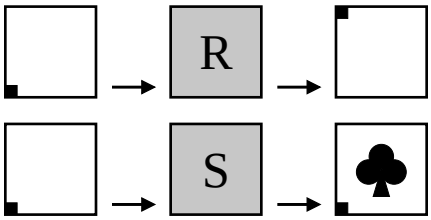
Warm-up: small steps first (no points)

W1. A square sheet has a black mark in the bottom left corner. You turn it 3 quarter turns clockwise. Where is the mark now?



A. top left ***B.*** in the middle ***C.*** bottom right
D. top right ***E.*** bottom left

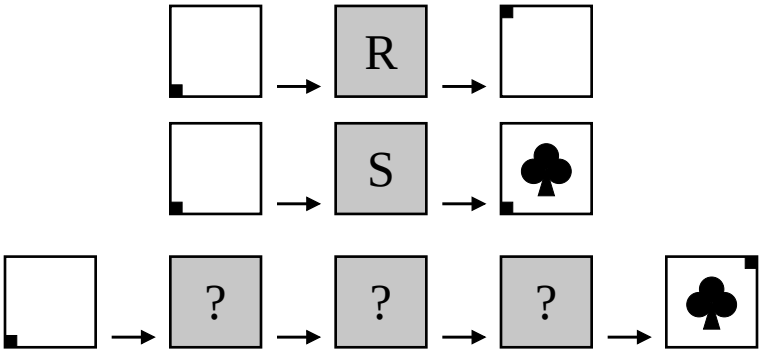
W2. Machine R turns a square sheet of paper a quarter turn clockwise. Machine S prints a club in the middle (see the pictures). The sheet goes through **R S R R**. How many quarter turns does the club make?



A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

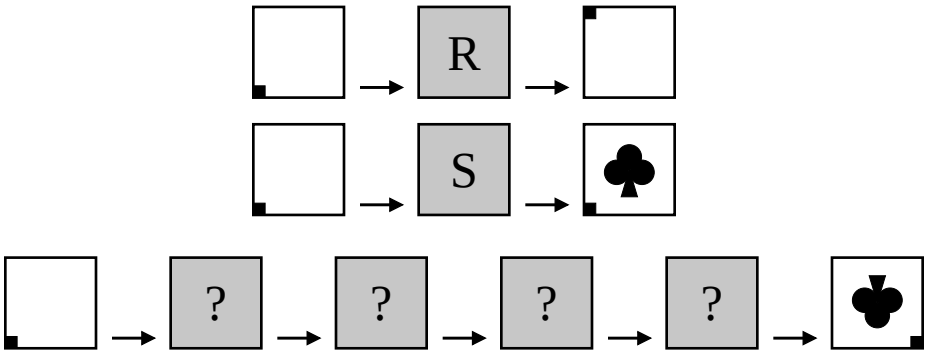
Part A: Each question is worth 3 points

1. Machine R turns a square sheet of paper a quarter turn clockwise. Machine S prints a club in the middle (see the pictures). Anna uses the machines 3 times in a row and gets the picture at the end of the bottom row. In which order does Anna use the machines?



A. SSS ***B.*** RRS ***C.*** SRR ***D.*** RSS ***E.*** SRS

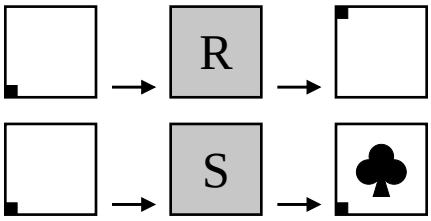
2. Machine R turns a square sheet of paper a quarter turn clockwise. Machine S prints a club in the middle (see the pictures). Tom uses the machines 4 times in a row and gets the picture at the end of the bottom row. In which order does Tom use the machines?

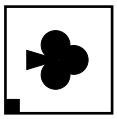


A. SSRS ***B.*** SRRR ***C.*** RSRR ***D.*** RRSR ***E.*** SSRR

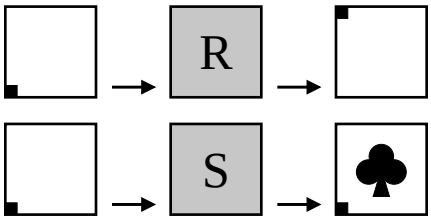
Part B: Each question is worth 4 points

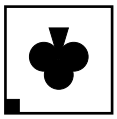
3. Machine R turns a square sheet of paper a quarter turn clockwise. Machine S prints a club in the middle (see the pictures). Anna puts a sheet in the machines in this order: **R R S R**. What does the sheet look like at the end?



A. ***B.***  ***C.*** ***D.*** ***E.***

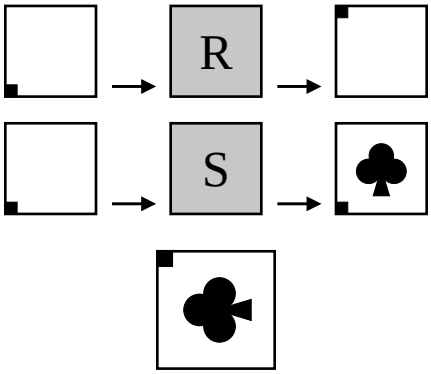
4. Machine R turns a square sheet of paper a quarter turn clockwise. Machine S prints a club in the middle (see the pictures). Sara puts a sheet in the machines in this order: **R R S R R**. What does the sheet look like at the end?



A. ***B.***  ***C.*** ***D.*** ***E.***

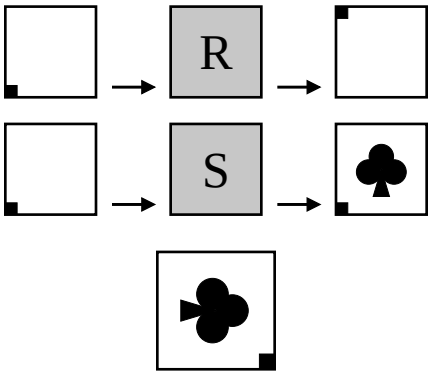
Part C: Each question is worth 5 points

5. Machine R turns a square sheet of paper a quarter turn clockwise. Machine S prints a club in the middle (see the pictures). Tom uses machine S once and machine R as often as needed. How many times at least must Tom use machine R to get this picture?



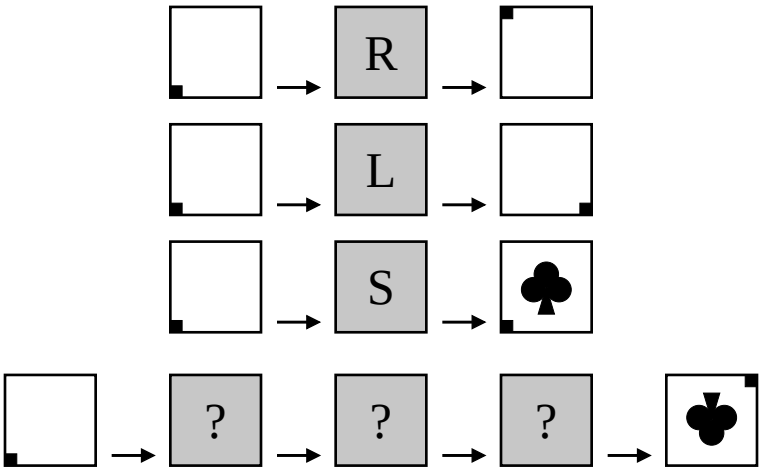
A. 1 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 9

6. Machine R turns a square sheet of paper a quarter turn clockwise. Machine S prints a club in the middle (see the pictures). Mia uses machine S once and machine R as often as needed. How many times at least must Mia use machine R to get this picture?



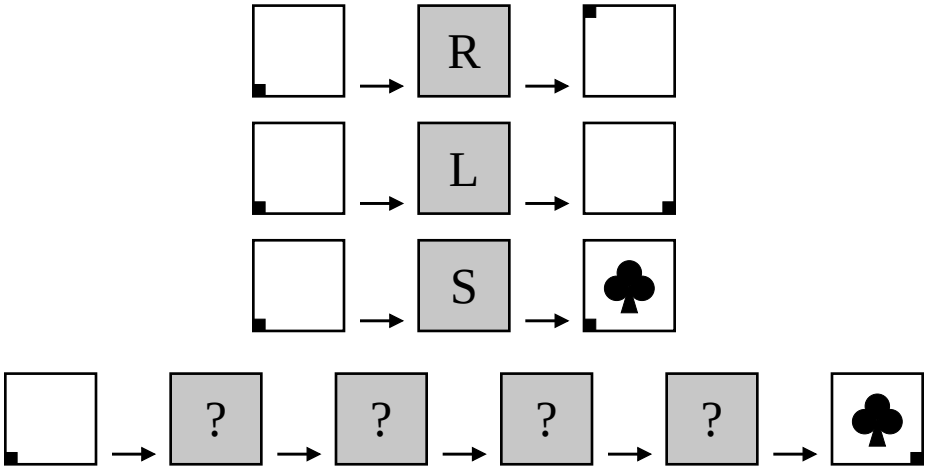
A. 3 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 7

7. Machine R turns a square sheet of paper a quarter turn clockwise. Machine S prints a club in the middle (see the pictures). Machine L turns the sheet a quarter turn anticlockwise. Leo uses the machines 3 times in a row and gets the picture at the end of the bottom row. Which order works?



A. LSL ***B.*** SRR ***C.*** SRL ***D.*** SSR ***E.*** LLS

8. Machine R turns a square sheet of paper a quarter turn clockwise. Machine S prints a club in the middle (see the pictures). Machine L turns the sheet a quarter turn anticlockwise. Else uses the machines 4 times in a row and gets the picture at the end of the bottom row. Which order works?



A. LLSR ***B.*** SLRR ***C.*** SRLL ***D.*** LLRS ***E.*** RRSR

Answer Key

W1.	**C**	**W2.**	**C**	**1.**	**B**	**2.**	**C**	**3.**	**E**	**4.**	**B**
5.	**D**	**6.**	**A**	**7.**	**B**	**8.**	**D**				

Solutions

W1. C (bottom right) — Clockwise the mark goes: bottom left, top left, top right, bottom right. [set-01/by_level/Grade_03-04_Ecolier/2023 Q22 · w_where_mark]

W2. C (2) — The club only turns with the R after S: 2. [set-01/by_level/Grade_03-04_Ecolier/2023 Q22 · w_club_after]

1. B (RRS) — The corner mark turned 2 quarter turns, so machine R is used 2 times. The club stands upright, so S comes last: RRS. [set-01/by_level/Grade_03-04_Ecolier/2023 Q22 · which_order]

2. C (RSRR) — The corner mark turned 3 quarter turns, so machine R is used 3 times. The club turned 2 quarter turns, so 2 R come after S: RSRR. [set-01/by_level/Grade_03-04_Ecolier/2023 Q22 · which_order]

3. E — The mark turns with every R: 3 quarter turns. The club is printed upright and then turns only with the 1 R after S. [set-01/by_level/Grade_03-04_Ecolier/2023 Q22 · which_picture]

4. B — The mark turns with every R: 4 quarter turns. The club is printed upright and then turns only with the 2 R after S. [set-01/by_level/Grade_03-04_Ecolier/2023 Q22 · which_picture]

5. D (5) — The club is turned 3 quarter turns, so 3 R come after S. The mark needs 2 more quarter turns before S. Together 5. [set-01/by_level/Grade_03-04_Ecolier/2023 Q22 · fewest_turns]

6. A (3) — The club is turned 1 quarter turn, so 1 R comes after S. The mark needs 2 more quarter turns before S. Together 3. [set-01/by_level/Grade_03-04_Ecolier/2023 Q22 · fewest_turns]

7. B (SRR) — Check each order: follow the corner mark through every machine and turn the club only with the machines after S. Only SRR gives the picture. [set-01/by_level/Grade_03-04_Ecolier/2023 Q22 · left_machine]

8. D (LLRS) — Check each order: follow the corner mark through every machine and turn the club only with the machines after S. Only LLRS gives the picture. [set-01/by_level/Grade_03-04_Ecolier/2023 Q22 · left_machine]